

Hackerrank Website analysis

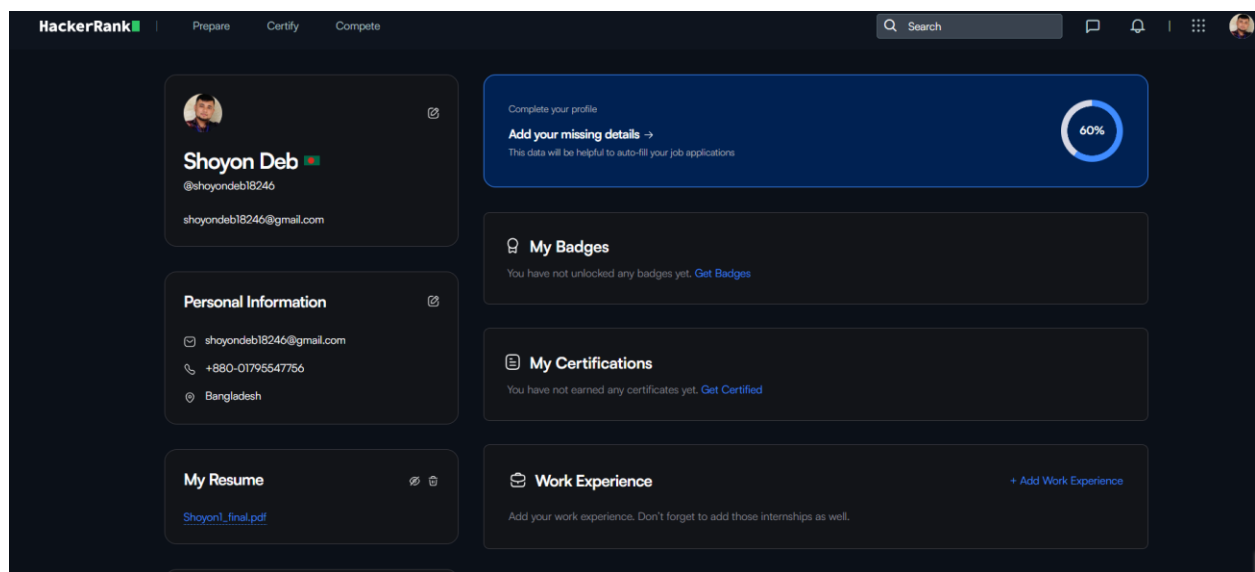
Overview

HackerRank is a platform for practicing coding, preparing for interviews, and connecting with employers. It offers coding challenges, skill assessments, contests, and hiring solutions for companies. Unlike Codeforces (contest-heavy) or LeetCode (interview-heavy), HackerRank balances coding practice + hiring integration.

Contestants or user panel:

Profile section:

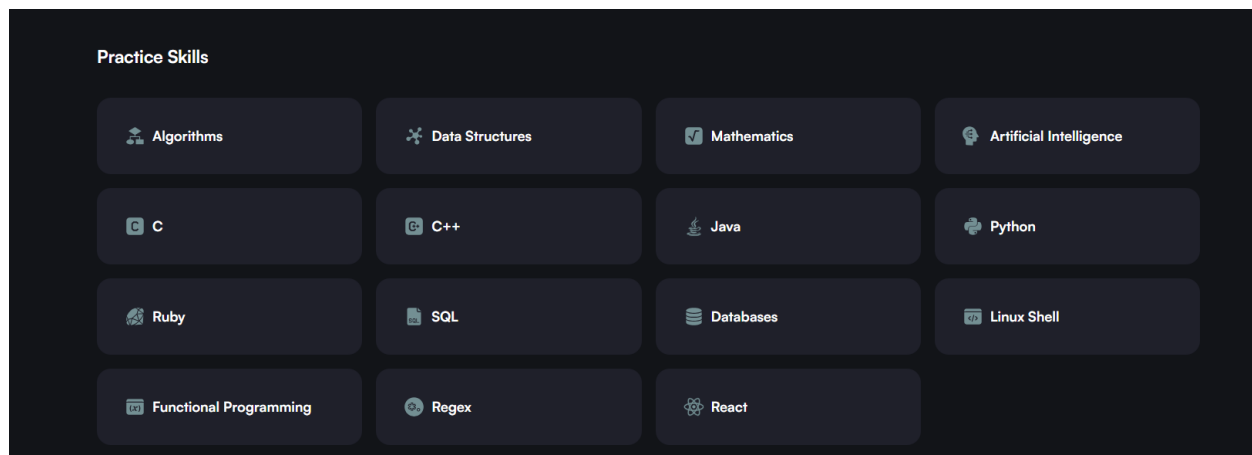
All the achievements will be shown on the profile section along with other information.



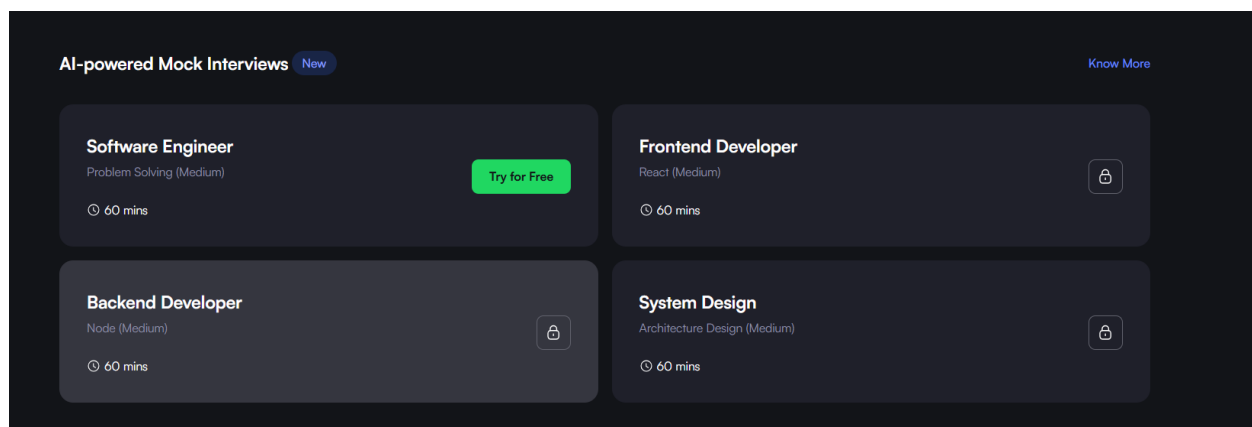
Key Features

1. Problem Sets & Domains

- Covers Algorithms, Data Structures, Mathematics, AI, Databases, SQL, Functional Programming, and more.
- Problems are tagged by domain and difficulty (Easy, Medium, Hard).
- Focus on practical coding skills often needed in industry.

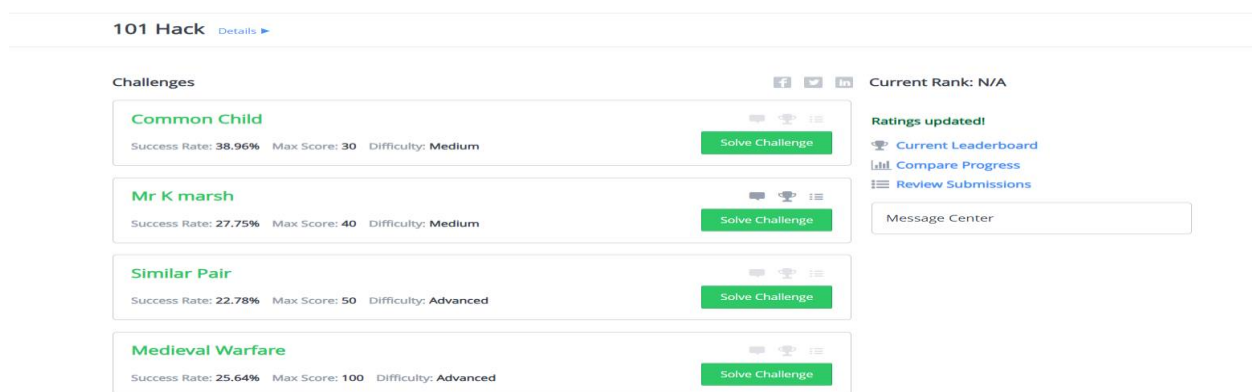


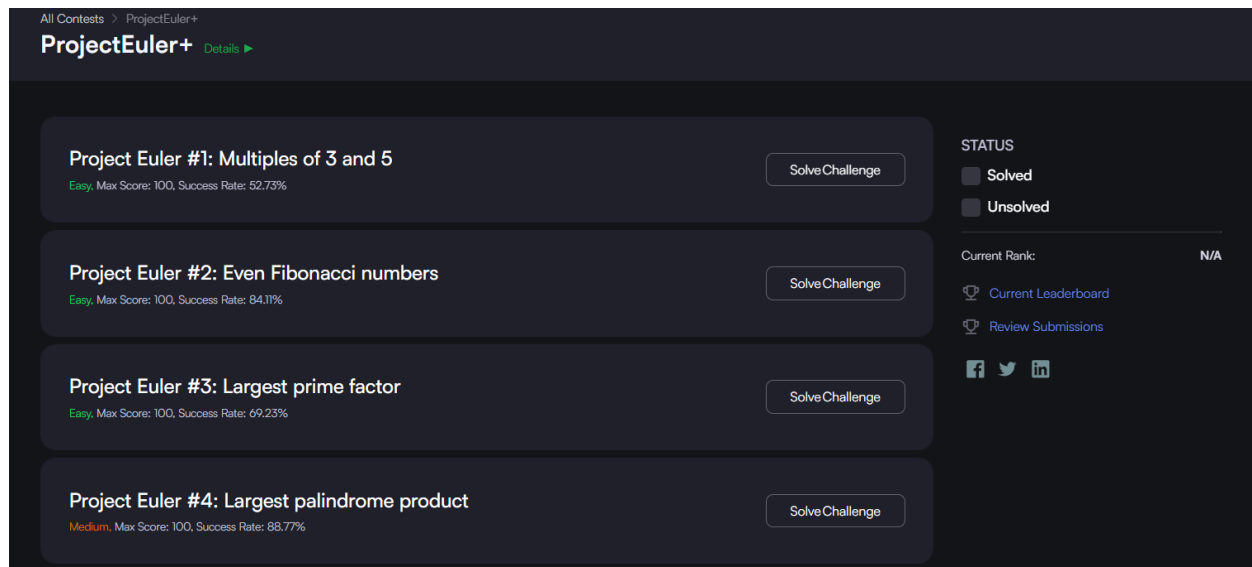
- AI Powered mock interview.



2. Contests

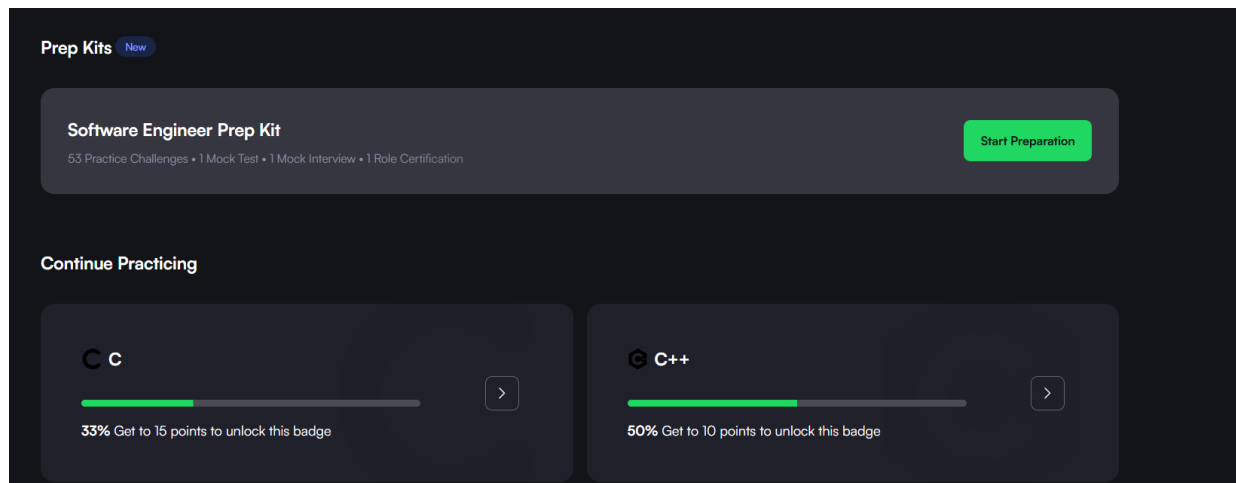
- Regular Contests like 101 Hack, Week of Code, Women's Cup, and College Contests.
- Competitive Hiring Contests where companies recruit directly from rankings.
- Leaderboards for global and regional performance.





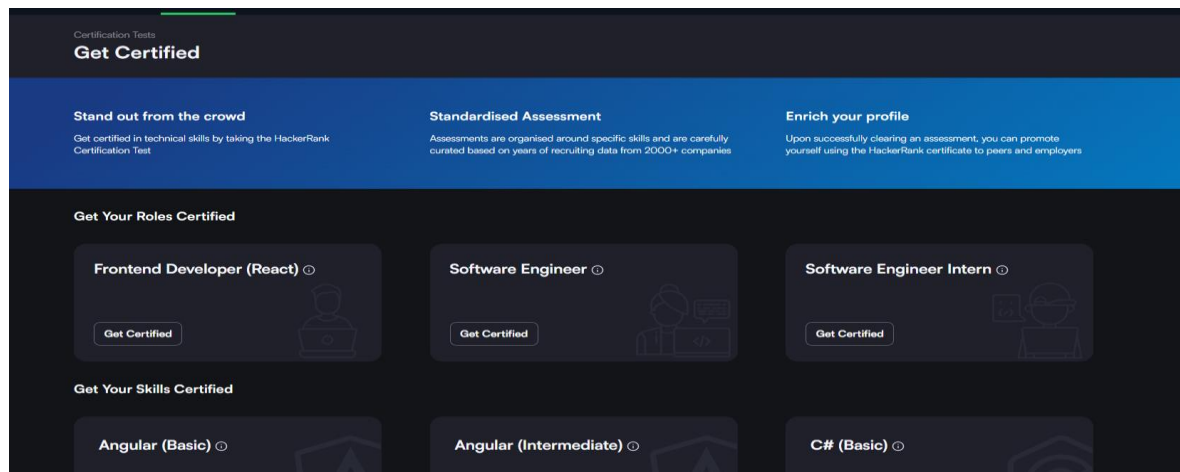
3. Interview Preparation

- Interview Preparation Kit: curated problem sets for coding interviews.
- Company-Specific Questions: problems frequently asked in companies (Google, Microsoft, Amazon, etc.).
- Mock Interviews with real-time coding environment.



4. Skills Certifications

- Users can take HackerRank Skill Certifications in Data Structures, Python, Java, SQL, Problem Solving, etc.
- Certificates can be added to LinkedIn to attract employers.



5. Employer / Recruiter Integration

- Employers can create coding tests and invite candidates.
- Recruiters get reports with code, runtime, and plagiarism checks.
- Virtual interview environment with code editor, chat, and whiteboard.

Create Contest

Host your own coding contest on HackerRank. You can practice and compete with friends from your organization or school. Select from our library of over 1,500 coding challenges or create your own.

Get started by providing the initial details for your contest.

Contest Name *

Start Time * at BDT [?](#)

End Time * at BDT [?](#)

☐ This contest has no end time.

Organization Type *

Organization Name *

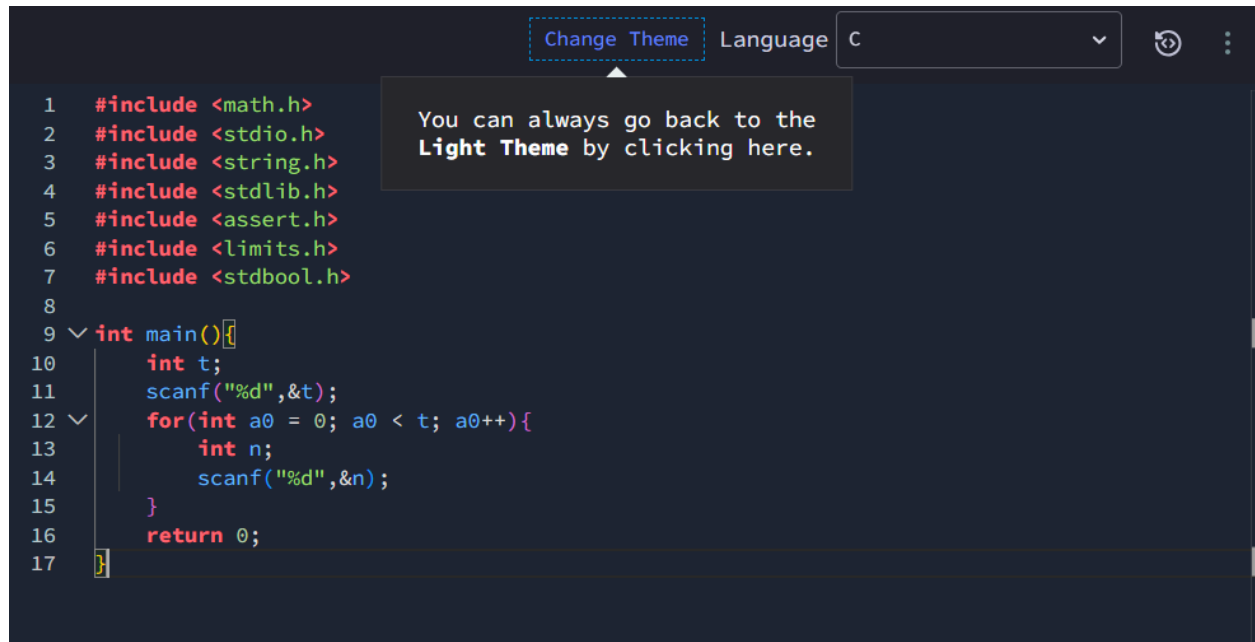
[Get Started](#)

6. Community Features

- Discussion forums for each problem.
- Interview experience sharing blogs.
- Contests for universities and groups.

User Interface & Experience

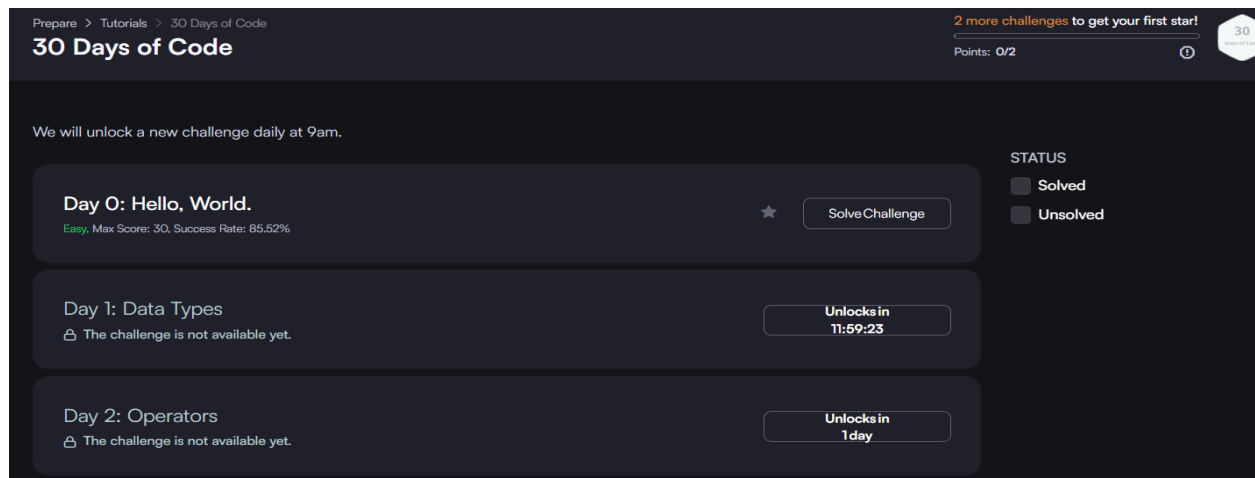
- Modern & Simple UI: Easy navigation across problem sets, contests, and interview prep.
- Online IDE: Built-in editor supports multiple programming languages.



- Custom Test Cases: Run solutions with your own inputs.
- Gamification: Badges, points, leaderboards, and certificates motivate learners.
- Mobile-Friendly: Responsive design.

Strengths

- Strong industry integration (companies actually hire via HackerRank).
- Wide domain coverage (not just algorithms, but also SQL, AI, functional programming, etc.).
- Free certifications that boost employability.
- Good for beginners with guided learning paths (30 Days of Code).



- Contests + career opportunities in one platform.

Weaknesses

- Contest frequency is lower than Codeforces.
- Problem discussions are less active compared to LeetCode.
- Some problems are too straightforward, focusing on syntax rather than deep problem-solving.

HackerRank Admin Panel Functionalities

The Admin Panel is used by HackerRank staff, recruiters, and problem setters to manage users, contests, and recruitment workflows.

User Management

- Manage user accounts (students, recruiters, companies).
- Approve, suspend, or ban accounts in case of violations.
- Provide recruiter/admin roles to company representatives.

Problem Management

- Add new problems with:
 - Statement, input/output format, constraints, examples.
 - Difficulty level and domain (DSA, SQL, Python, etc.).
 - Tags and company relevance (e.g., “Amazon interview question”).
- Upload hidden test cases (sample + system).
- Update/edit problem statements and editorial solutions.

Contest & Test Management

- Create public contests (like “Week of Code”) or private tests for companies.
- Configure test settings:
 - Contest type (practice vs hiring test).
 - Duration, start/end time.
 - Languages allowed.
- Assign problems from the problem bank or create new ones.
- Monitor live contests/tests (submission stats, cheating detection).

Recruiter Tools

- Create custom coding assessments for hiring.
- Add MCQs (theory + aptitude questions) in addition to coding problems.
- Invite candidates via email.
- Generate candidate reports (scores, execution time, plagiarism check).

Platform Management

- Announcements and news updates.
- Monitor platform usage, server load, and reports.
- Manage certifications

HackerRank Judge Panel Functionalities

The **Judge Panel** is the automated backend system that evaluates submissions in both **practice** and **recruitment tests**.

Step 1: Submission Intake

- Receive code submissions from candidates/participants.
- Identify the chosen programming language.
- Queue the submission for processing.

Step 2: Code Compilation

- Compile the submitted code in a sandboxed environment.
- Catch Compilation Errors early and report them back.

Step 3: Execution on Test Cases

- Run code against multiple **test case groups**:
 - Sample Test Cases (visible to user).
 - Hidden Test Cases (to prevent hardcoding).

- Stress & Edge Test Cases (extreme input sizes).
- Track runtime and memory usage.

Sample Input 0

```
2
10
100
```

Sample Output 0

```
23
2318
```

Explanation 0

For $N = 10$, if we list all the natural numbers below 10 that are multiples of 3 or 5, we get 3, 5, 6 and 9. The sum of these multiples is 23.

Similarly for $N = 100$, we get 2318.

Step 4: Verdict Assignment

- Generate result for each submission:
 - Accepted
 - Wrong Answer
 - Time Limit Exceeded (TLE)
 - Memory Limit Exceeded (MLE)
 - Runtime Error
 - Compilation Error
- In hiring tests: also calculate score percentage and mark partial correctness if some cases pass.

Step 5: Cheating & Plagiarism Check

- Detect code similarity between candidates.
- Monitor suspicious behaviors.
- Flag recruiters if plagiarism is detected.

Step 6: Reports & Ranking

- In contests: update leaderboard in real time.
- In hiring tests: generate detailed candidate report with:
 - Problem-wise performance.
 - Execution time.
 - Percentage score.
 - Plagiarism status.